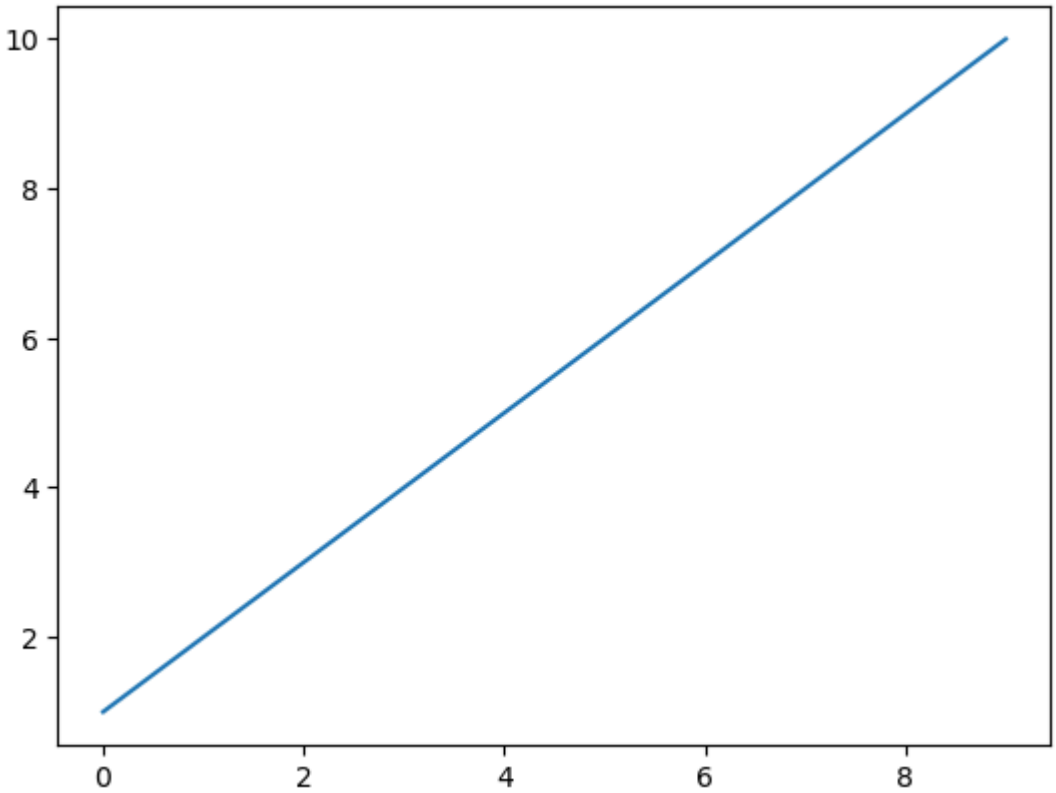
In [1]:
```
import matplotlib.pyplot as plt
```

1. plt (basics)

https://matplotlib.org/stable/plot_types/index.html

In [2]:
```
a = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

plt.plot(a)   #si l'axe des x n'est pas définit, ça va être: [i for i in range(l
```

Out[2]:
```
[<matplotlib.lines.Line2D at 0x20a041cca60>]
```

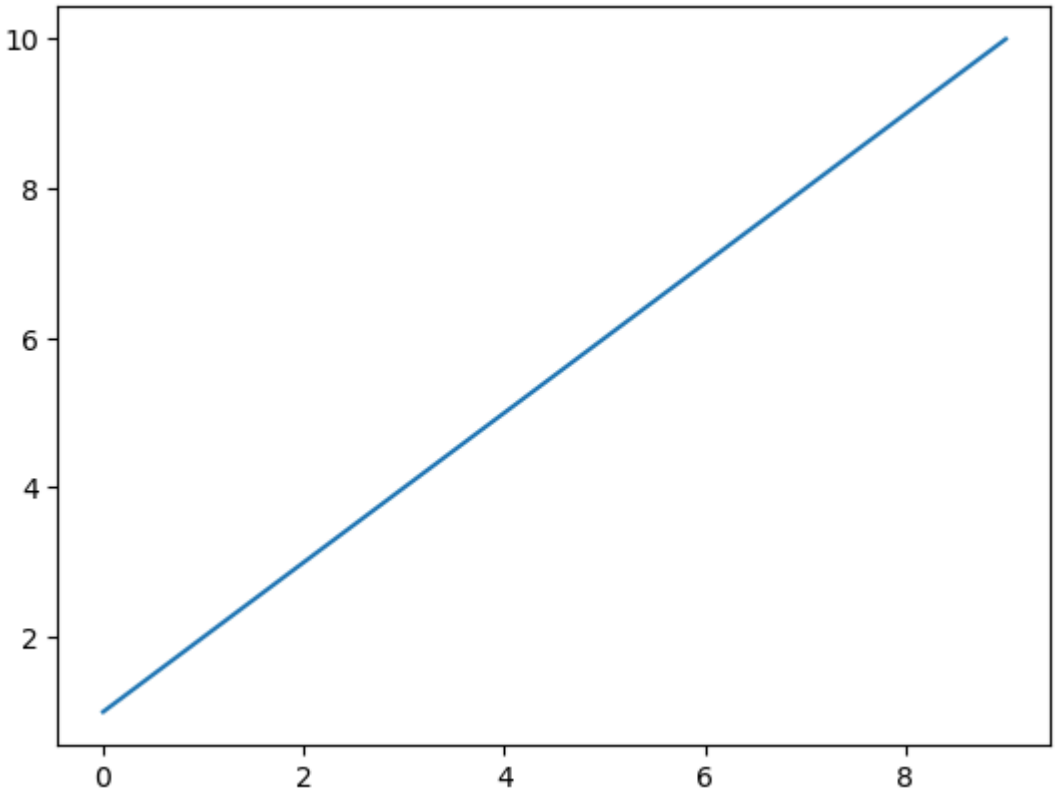


In [3]:
```
import numpy as np  #pour plus de rapidité
```

In [4]:
```
a = np.array(a)
```

In [5]:
```
plt.plot(a)
```

Out[5]:
```
[<matplotlib.lines.Line2D at 0x20a041816f0>]
```

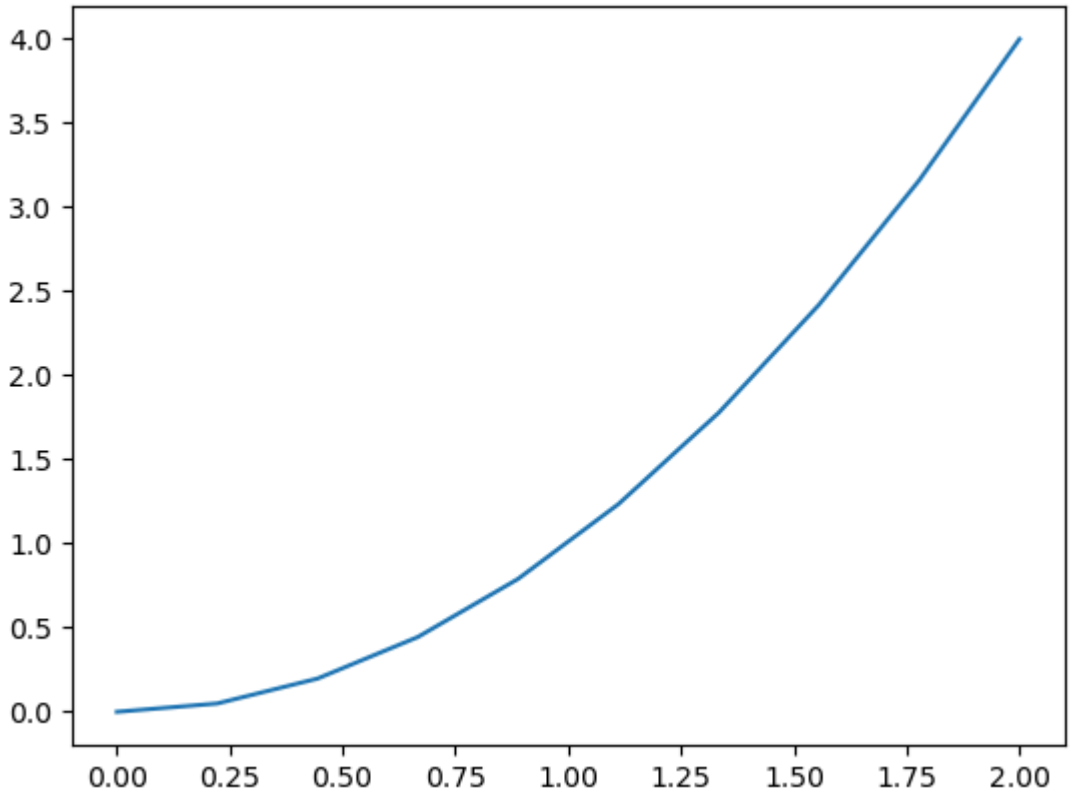


In [ ]:

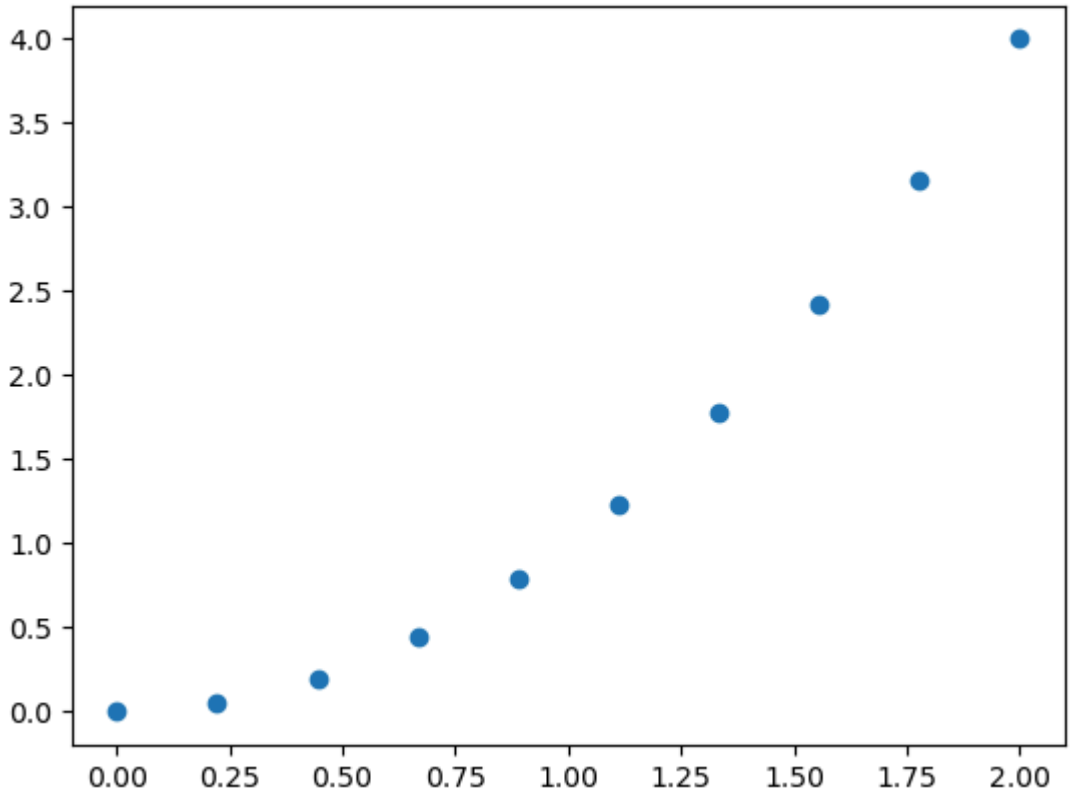
In [7]:
```
X = np.linspace(0, 2, 10)
X
```

Out[7]:
```
array([0.        , 0.22222222, 0.44444444, 0.66666667, 0.88888889,
       1.11111111, 1.33333333, 1.55555556, 1.77777778, 2.        ])
```

In [9]:
```
X = np.linspace(0, 2, 10)
y = X**2

plt.plot(X, y)       #x and y now
# plt.xticks(X)      #Set the x-axis ticks to match the values in X
plt.show()
```



In [10]:
```
plt.scatter(X, y)
plt.show()
```



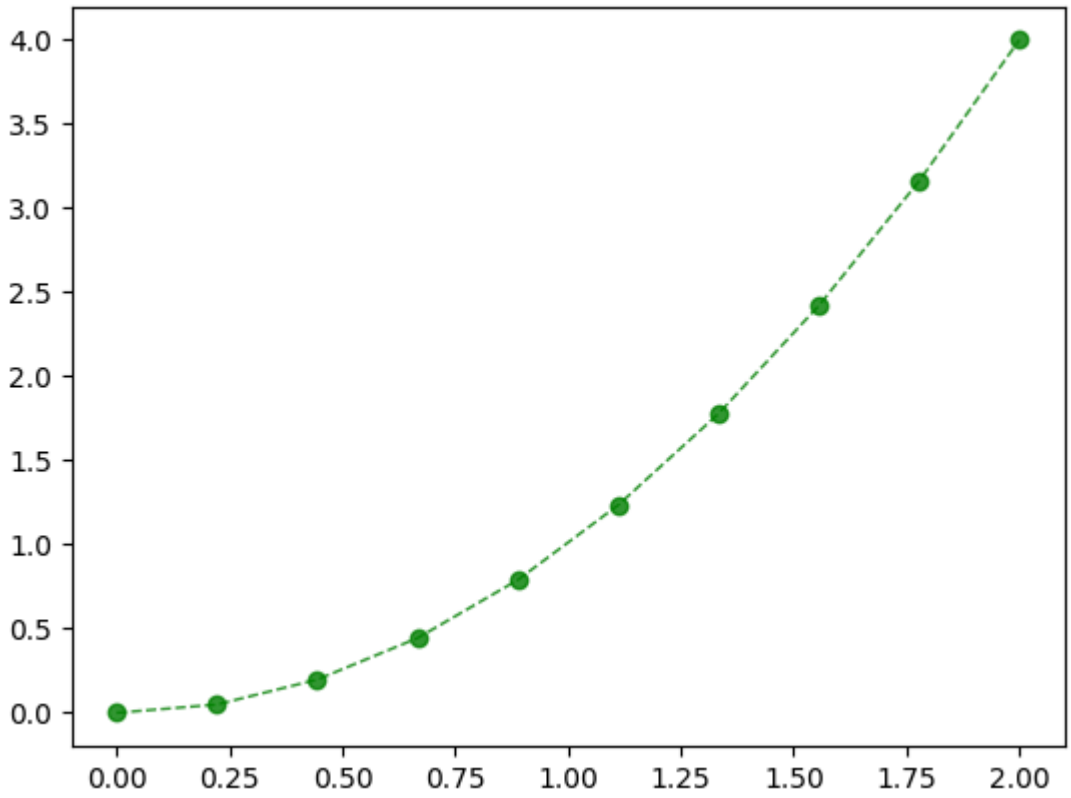
Styles Graphiques

Il existe beaucoup de styles a ajouter aux graphiques. Voici les plus importants a retenir :

- **c** : couleur de la ligne

- **lw** : epaisseur de la ligne (pour les graphiques plot)
- **ls** : style de la ligne (pour les graphiques plot)
- **size** : taille du point (pour les graphiques scatter **seul**)
- **marker** : style de points (pour les graphiques scatter)
- **alpha** : transparence du graphique

In [1]:
```
import matplotlib.pyplot as plt
import numpy as np

X = np.linspace(0, 2, 10)
y = X**2

plt.plot(X, y, c='green', lw=1, ls='--', marker='o', alpha=0.8)  #D : diamond
plt.show()
```



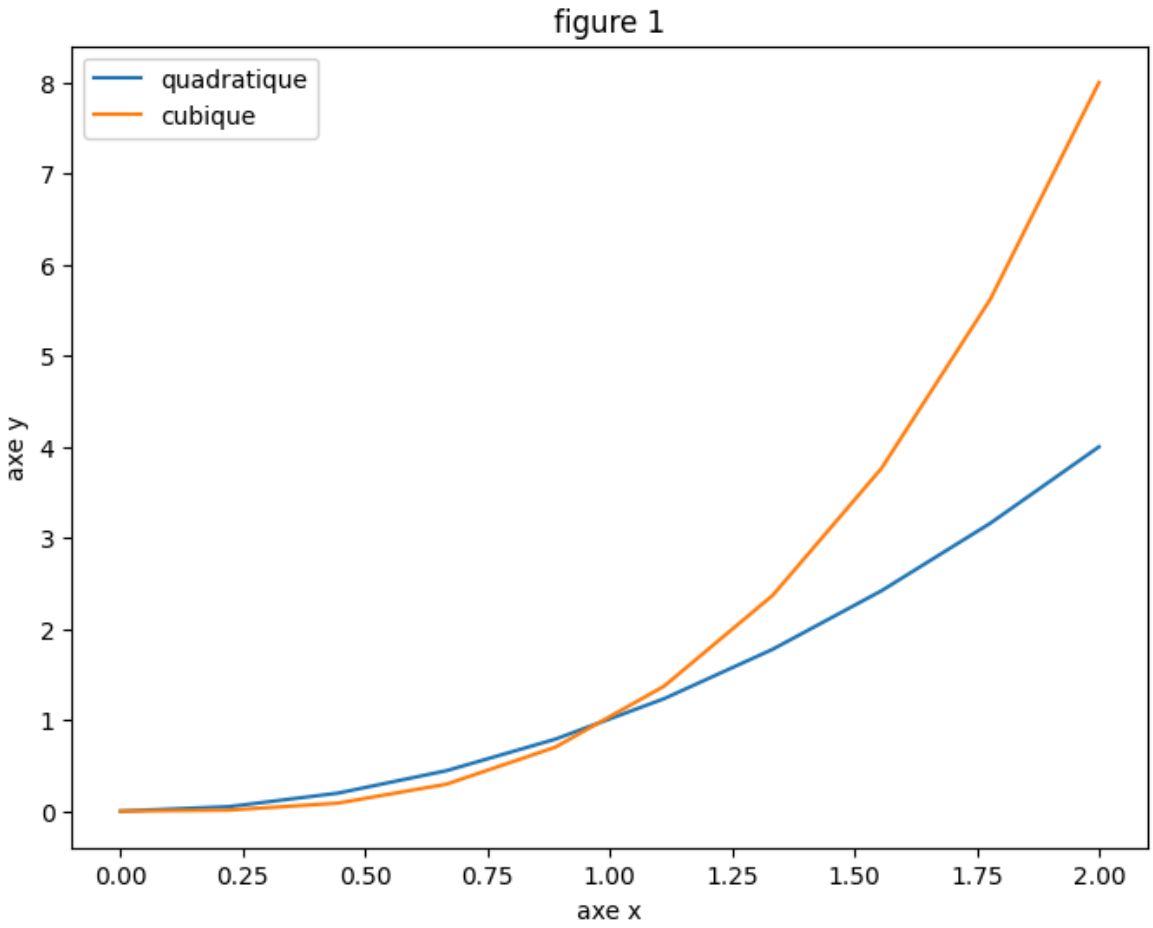
2. Cycle de vie d'une figure

Pour créer des figures proprements, on doit suivre le cycle de vie suivant :

1. **plt.figure**(figsize=(a,b))
2. **plt.plot()**
3. optional **Extras (titre, axes, legendes)**
4. optional **plt.save()**
5. **plt.show()**

In [15]:
```
X = np.linspace(0, 2, 10)

plt.figure(figsize=(8, 6)) # Création d'une figure
plt.plot(X, X**2, label='quadratique') # premiere courbe
plt.plot(X, X**3, label='cubique') # deuxieme courbe
```

```
# Autres informations
plt.title('figure 1') # titre
plt.xlabel('axe x') # axes
plt.ylabel('axe y') # axes
plt.legend() # legend, takes what's in label

# plt.savefig('figure.png') # sauvegarde la figure dans le repertoire de travail
plt.show() # affiche la figure
```



3. Subplots

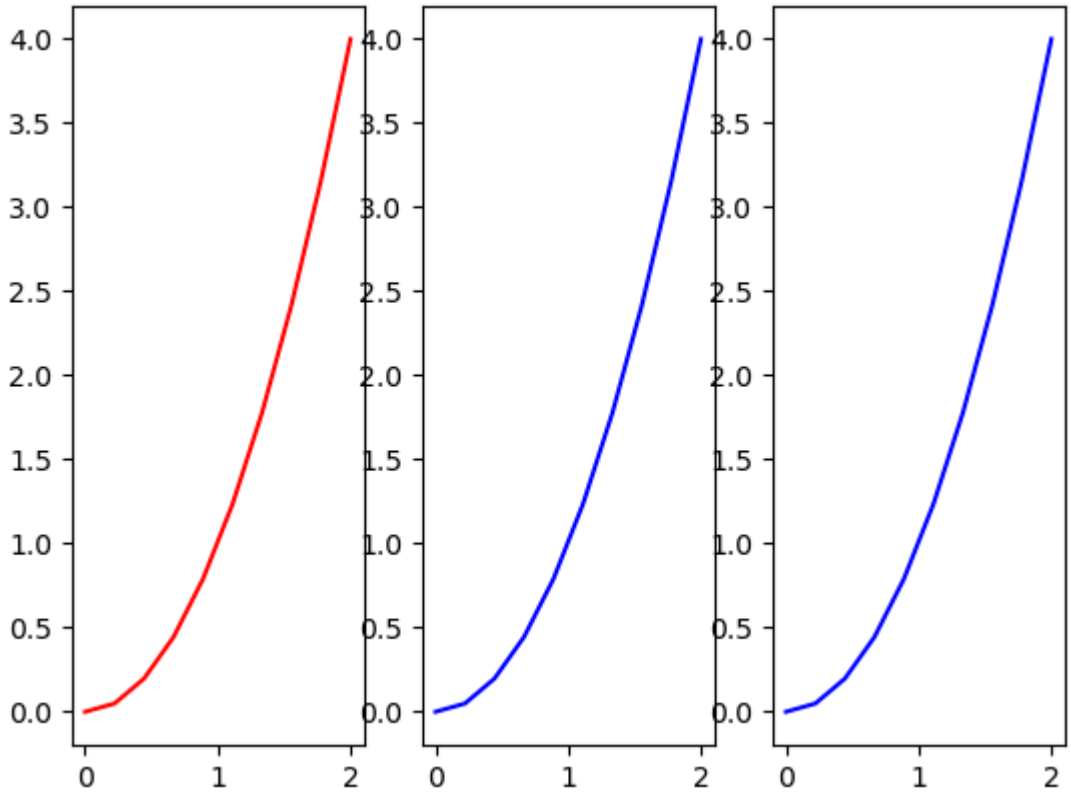
Les subplots sont un autre éléments a ajouter pour créer plusieurs graphiques sur une meme figure

In [14]:

```
        #Ligne, Col, idx

plt.subplot(1, 3, 1)
plt.plot(X, y, c='red')
plt.subplot(1, 3, 2)
plt.plot(X, y, c='blue')
plt.subplot(1, 3, 3)
plt.plot(X, y, c='blue')

# plt.subplot(2, 2, 3)
# plt.plot(X, y, c='red')
# plt.subplot(2, 2, 4)
# plt.plot(X, y, c='blue')
```

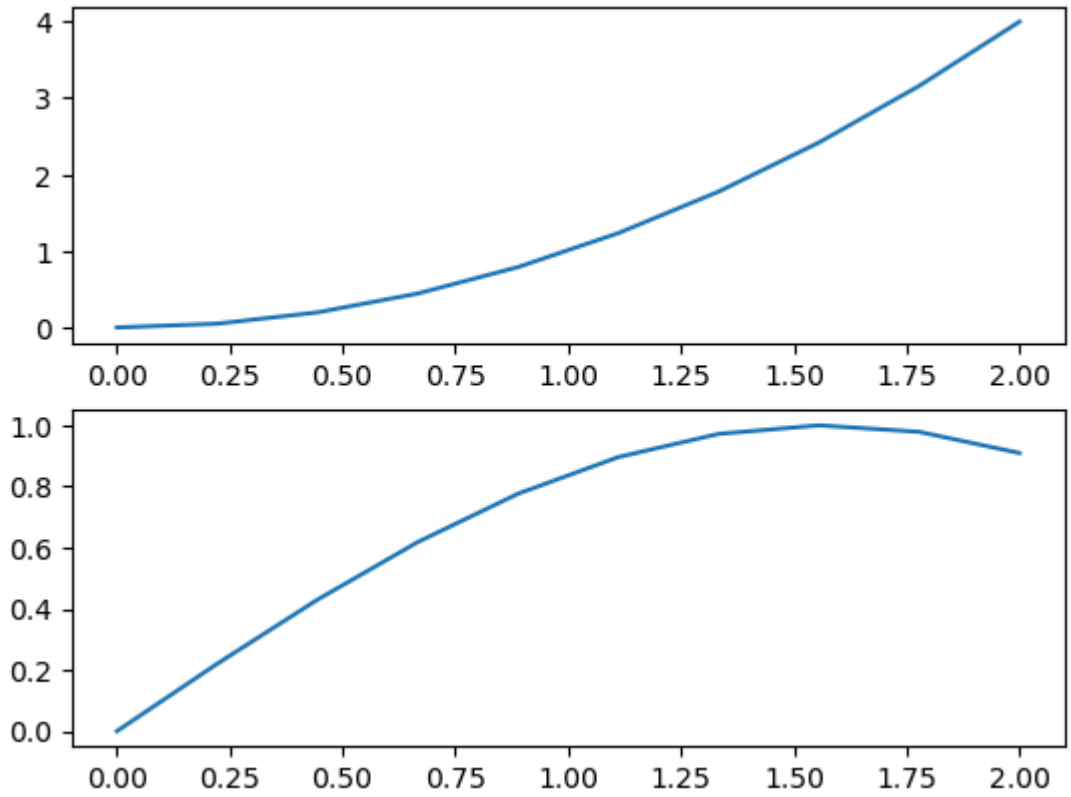
Out[14]:

```
[<matplotlib.lines.Line2D at 0x20a042d2a40>]
```



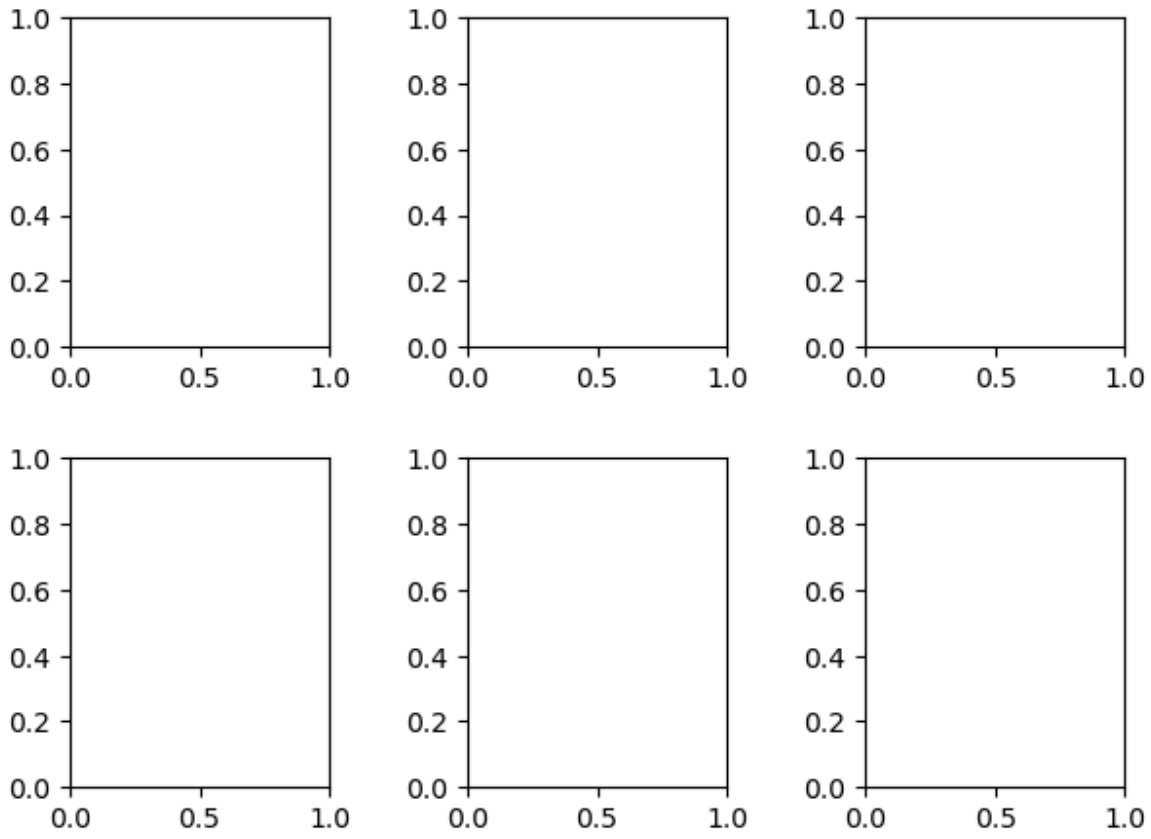
subplot using classes

In [27]:
```
fig, ax = plt.subplots(2, 1, sharex=False) #ou sharex=True partage du meme axe p
ax[0].plot(X, y)
ax[1].plot(X, np.sin(X))
plt.show()
```



In [ ]:

In [30]:
```
fig, axs = plt.subplots(2, 3)   #2 lignes, 3 collonnes
plt.tight_layout(
    pad = 2, #area of the entire space
    h_pad = 2,  #vertical space betwee, figs
    w_pad = 2   #horizontal space between figs
    )
plt.show()
```



In [ ]:

4. les différents plots

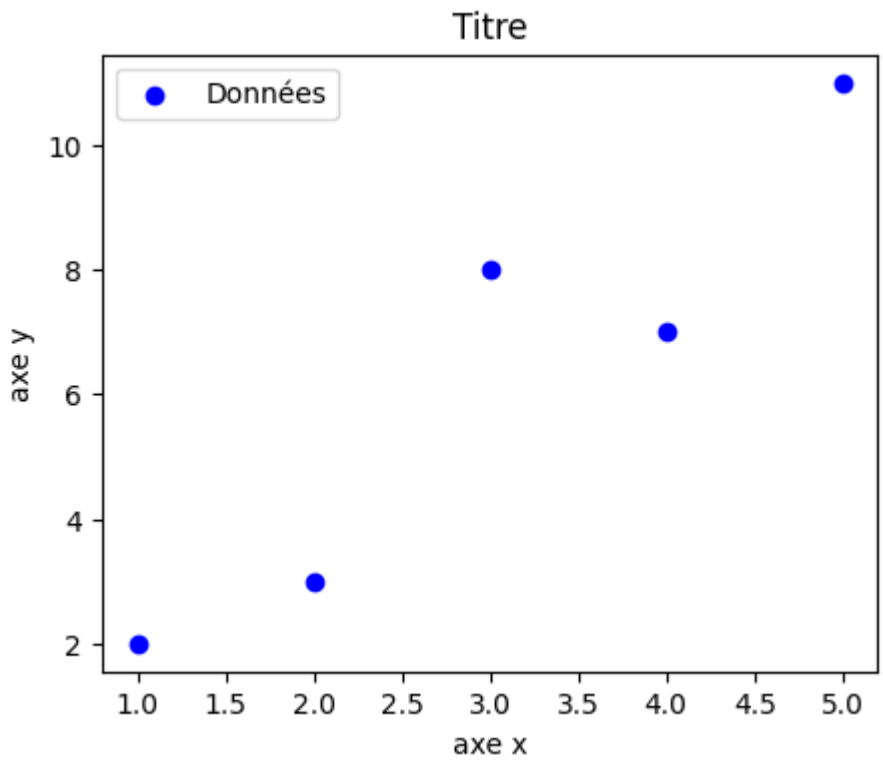
en plus du line plot

4.1. Scatter plot

- Doc: https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.scatter.html

In [1]:
```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [2, 3, 8, 7, 11]

plt.figure(figsize=(5, 4))

plt.scatter(x, y, label='Données', color='blue', marker='o')  #marker= o, *, ^,

plt.xlabel('axe x')
plt.ylabel('axe y')
```

```
plt.title('Titre')
plt.legend()
plt.show()
```



4.3 Histogram

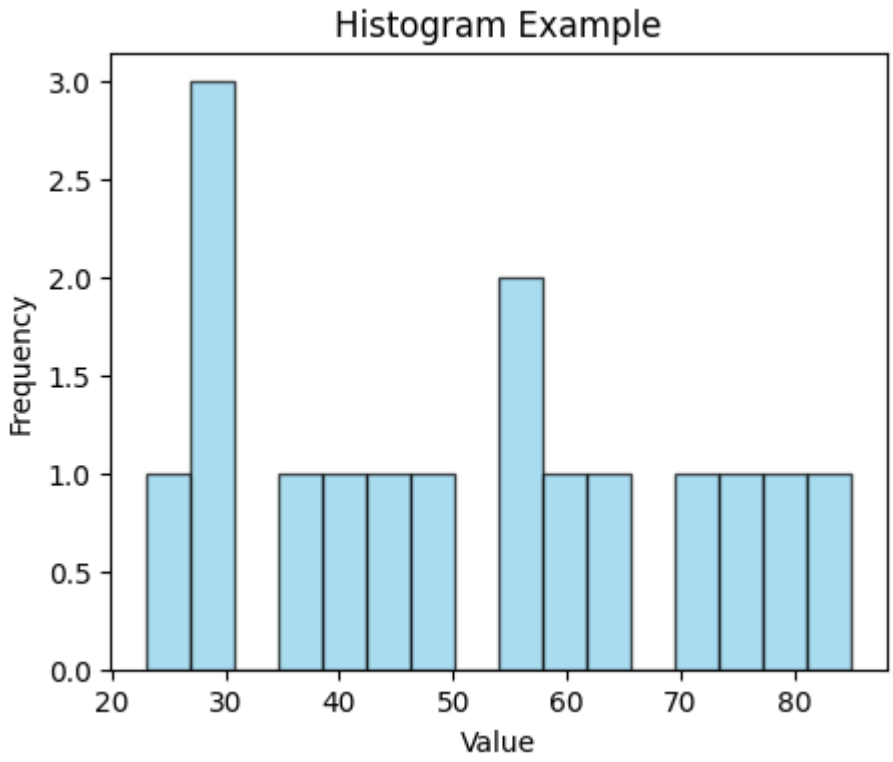
distribution discontinue des données

- Pour chaque valeur ou catégories --> combien elle a été répétée

In [34]:
```
## 30 se répéte trois fois, 55 se répéte deux fois

data = [23, 30, 30, 30, 37, 40, 45, 50, 55, 55, 60, 65, 70, 75, 80, 85]

len(data)
```

Out[34]: 16

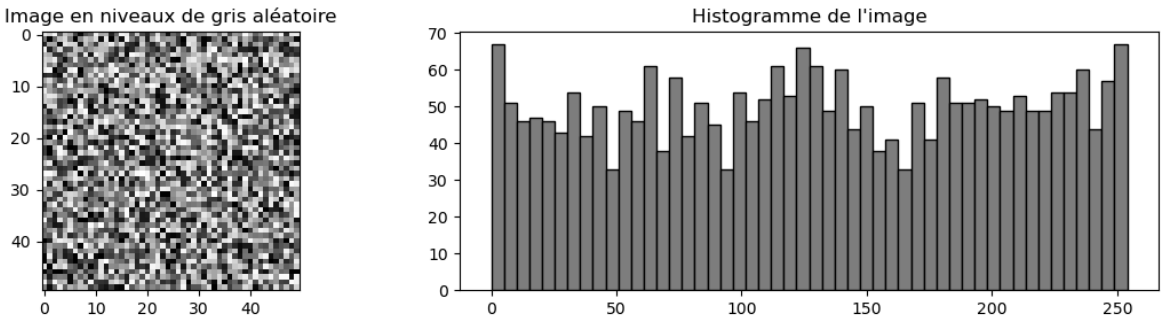
In [35]:
```
plt.figure(figsize=(5, 4))

plt.hist(data, bins=len(data),   ## bins : combien de catégories pour diviser la
         edgecolor='black',
         alpha=0.7, color='skyblue')

plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Histogram Example')
plt.show()
```



In [7]:
```
import numpy as np
import matplotlib.pyplot as plt

X = np.random.randint(0, 255, [50, 50])
print(np.max(X), np.min(X))
```

```
254 0
```

In [8]:
```
# Afficher l'image
plt.figure(figsize=(12, 3))
plt.subplot(1, 2, 1)
plt.imshow(X, cmap='gray')
plt.title('Image en niveaux de gris aléatoire')

# Afficher l'histogramme de l'image
plt.subplot(1, 2, 2)
plt.hist(X.ravel(), bins=50, color='gray', edgecolor='black')
plt.title('Histogramme de l\'image')

plt.tight_layout()
plt.show()
```



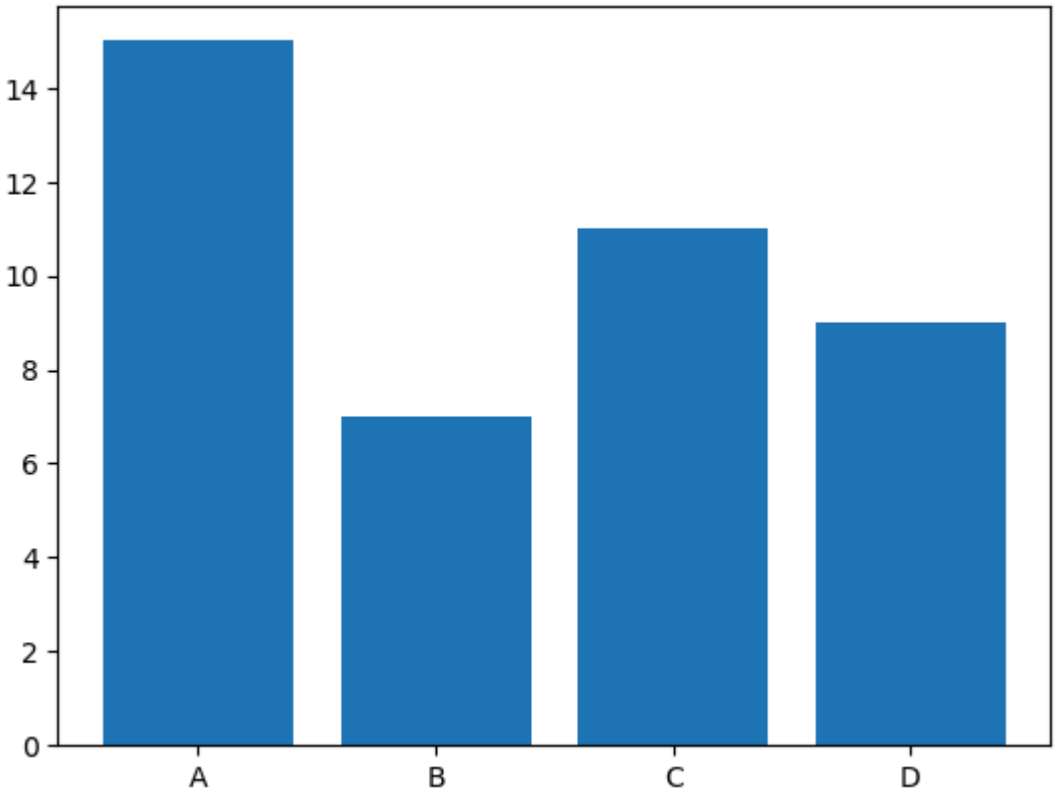
In [ ]:

Bar plot

In [9]:
```
import matplotlib.pyplot as plt
import pandas as pd

example = {'Category': ['A', 'B', 'C', 'D'],
        'Value': [15, 7, 11, 9]}

df_ex = pd.DataFrame(example)
df_ex
```

Out[9]:

	Category	Value
0	A	15
1	B	7
2	C	11
3	D	9

In [10]:
```
#on donne deux colonnes
        #les catégories  , les valeurs
plt.bar(df_ex['Category'], df_ex['Value'])
```

Out[10]: <BarContainer object of 4 artists>